

An Introduction to Vector Databases and Similarity Search

What Is a Vector Database

A vector database is a system designed to store, index, and search high-dimensional numerical vectors, typically produced by embedding models. Unlike traditional databases that search by exact match, vector databases search by similarity, returning the items whose vectors are geometrically closest to a query vector.

Distance Metrics

The three most common distance metrics used in vector search are cosine similarity, Euclidean distance (L2), and dot product. Cosine similarity measures the angle between two vectors and is unaffected by their magnitude, which makes it a popular default for text embeddings. Euclidean distance measures the straight-line distance between two points in vector space.

Approximate Nearest Neighbor Search

Exact nearest-neighbor search becomes computationally expensive as the number of vectors grows into the millions. To solve this, most vector databases use Approximate Nearest Neighbor (ANN) algorithms such as HNSW (Hierarchical Navigable Small World graphs) or IVF (Inverted File Index), which trade a small amount of accuracy for a large gain in query speed.

HNSW Explained

HNSW builds a multi-layered graph structure where the top layer contains few nodes with long-range connections, and each layer below contains progressively more nodes with shorter-range connections. A search starts at the top layer and greedily moves toward the query vector, descending layers until it reaches the bottom, where the nearest neighbors are found.

Chunking Strategy for RAG

When preparing documents for a vector database, text is typically split into overlapping chunks before embedding. A common starting point is a chunk size of 500 to 1000 characters with an overlap of 10 to 20 percent, which helps prevent relevant context from being split across chunk boundaries.

Popular Vector Database Options

Widely used vector databases include ChromaDB, which is lightweight and easy to self-host; Qdrant, which offers a generous free cloud tier; Pinecone, a fully managed commercial option; and Weaviate, which supports hybrid keyword and vector search out of the box.